

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2023 (2019 Scheme)

Course Code: CXT309**Course Name: OBJECT ORIENTED MODELING AND DESIGN**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|---|---|
| 1 | Differentiate between encapsulation and information hiding in object-oriented programming. | 3 |
| 2 | Explain the role of polymorphism in supporting dynamic dispatch in object-oriented programming. | 3 |
| 3 | Explain the role of events and states in dynamic modeling. | 3 |
| 4 | Describe the concept of nested state diagrams and their applications in dynamic modeling. | 3 |
| 5 | Explain the purpose of identifying object classes in object modelling. | 3 |
| 6 | What is the role of a data dictionary in object modeling? | 3 |
| 7 | What are the factors to be considered for the design of the algorithm by a designer? | 3 |
| 8 | What are the three basic approaches to implementing the dynamic model? | 3 |
| 9 | Explain Unified Modeling Language (UML). | 3 |
| 10 | Mention the purpose of use case diagram. | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|---|---|
| 11 | a) Discuss the concept of object-oriented development (OOD). | 7 |
| | b) What are the advantages of OOD over traditional procedural programming approaches. | 7 |

OR

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|----|---|----|
| 12 | a) Explain the concept of object modeling and its role in capturing and representing real-world requirements in software development. | 14 |
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Module -2

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|----|---|----|
| 13 | a) Discuss the concept of dynamic modeling and its significance in capturing and representing the behavioral aspects of software systems. | 14 |
|----|---|----|

OR

- 14 a) Explain the concept of state diagrams and their role in representing the behaviour of objects in object-oriented systems. 14

Module -3

- 15 a) Discuss the concept of object modeling and its role in capturing and representing the static structure of software systems. 14

OR

- 16 a) Explain the concept of dynamic modeling and the steps performed for constructing a dynamic model. 14

Module -4

- 17 a) You have been tasked with designing a game that involves multiple players interacting with each other in a virtual environment. How would you design the objects and associations necessary to manage the game state, player interactions, and scoring? What are some potential challenges you might encounter when designing these objects and associations? 14

OR

- 18 a) Discuss the various design optimization techniques used in object design and their role in enhancing system performance, scalability, maintainability, and reusability. 14

Module -5

- 19 a) Discuss the key notations, models, and concepts introduced by Grady Booch's object-oriented analysis and design methodology. 14

OR

- 20 a) Explain the concepts of actors and use-cases in Jacobson's methodology and their role in capturing the functional requirements of software systems. 14
